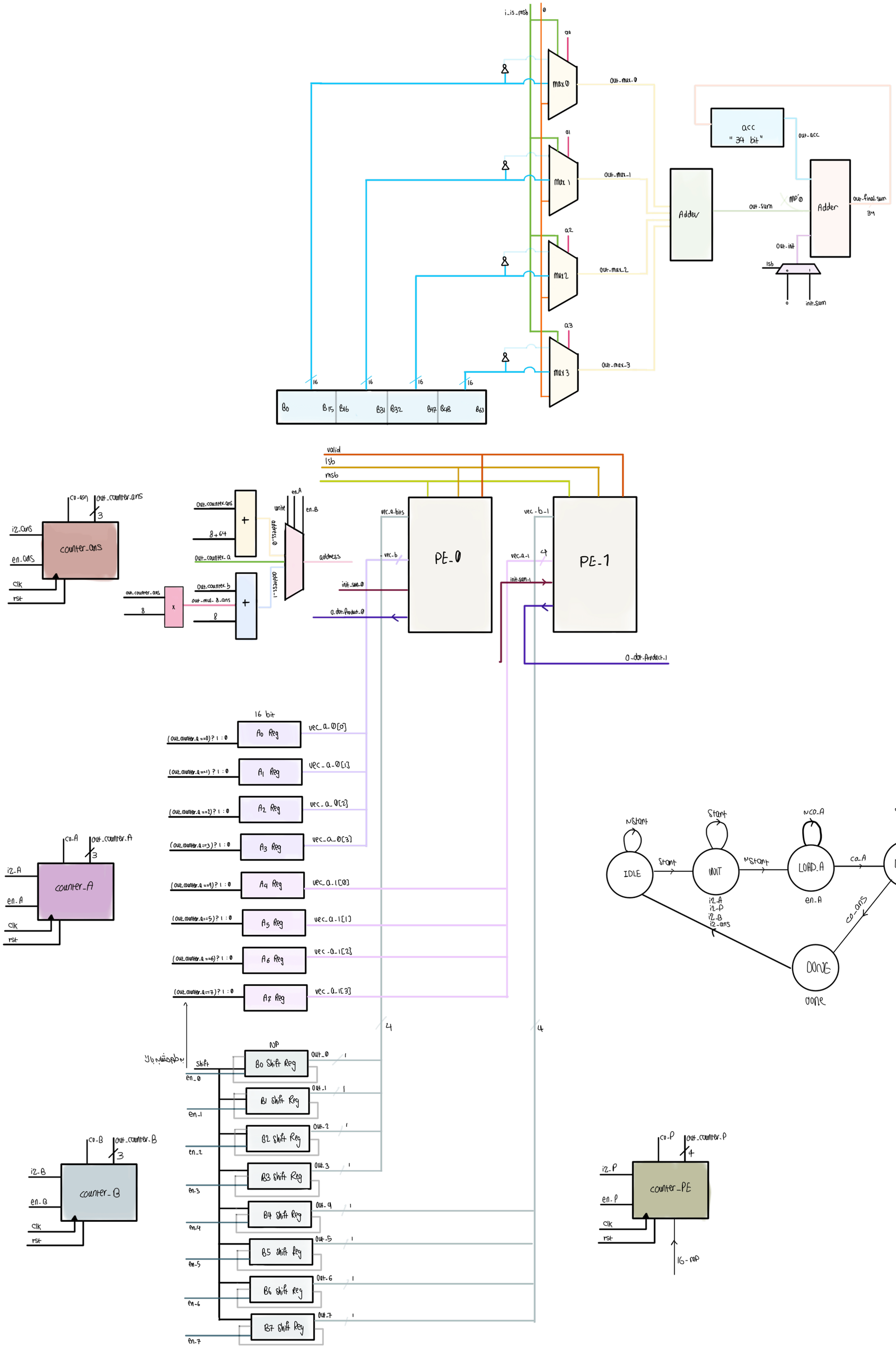

i_is_msb
0
a0
Mux 0
out_mux_0
a1
Mux 1
out_mux_1
a2
Mux 2
out_mux_2
a3
Mux 3
out_mux_3
16
16
16
16
B0
B15
B16
B31
B32
B47
B48
B63
Adder
acc
" 34 bit "
out_acc
Adder
out_sum
MP'0
out_final_sum
34
out_init
lsb
0
init_sum
valid
lsb
msb
PE_0
PE_1
vec_a_bits
vec_b
vec_b_1
vec_a_1
init_sum_0
init_sum_1
o_dot_product_0
o_dot_product_1
counter_ans
co_ans
i2_ans
en_ans
clk
rst
out_counter_ans
3
out_counter_a
out_counter_b
out_mul_a_ans
8
8_+_64
+
x
address
address_0
address_1
write
en_A
en_B
counter_A
co_A
out_counter_A
i2_A
en_A
counter_B
co_B
out_counter_B
i2_B
en_B
16 bit
A0 Reg
A1 Reg
A2 Reg
A3 Reg
A4 Reg
A5 Reg
A6 Reg
A7 Reg
(out_counter_a==0) ? 1 : 0
(out_counter_a==1) ? 1 : 0
(out_counter_a==2) ? 1 : 0
(out_counter_a==3) ? 1 : 0
(out_counter_a==4) ? 1 : 0
(out_counter_a==5) ? 1 : 0
(out_counter_a==6) ? 1 : 0
(out_counter_a==7) ? 1 : 0
vec_a_0[0]
vec_a_0[1]
vec_a_0[2]
vec_a_0[3]
vec_a_1[0]
vec_a_1[1]
vec_a_1[2]
vec_a_1[3]
4
4
MP
Shift
B0 Shift Reg
B1 Shift Reg
B2 Shift Reg
B3 Shift Reg
B4 Shift Reg
B5 shift Reg
B6 Shift Reg
B7 Shift Reg
en_0
en_1
en_2
en_3
en_4
en_5
en_6
en_7
out_0
out_1
out_2
out_3
out_4
out_5
out_6
out_7
counter_PE
co_P
out_counter_P
i2_P
en_P
16-MP
NStart
Start
NCO_A
NCO_B
NCO_P
IDLE
INIT
LOAD_A
LOAD_B
START MUL
MUL
END MUL
WRITING IN MEM
DONE
i2_A
i2_P
i2_B
i2_ans
en_A
en_B
i2_P
MSB is valid
Shift_B
en_P
Shift_b
en_P
is_valid
lsb
write
en_ans
done